

Week 3 - Power Query and Data Preparation

Power BI - The Insight Stack

What Power Query does

Power Query Editor: where raw data becomes report-ready data



Choose your preparation approach

Clean

Fix the data first

- ✓ Remove errors
- ✓ Fix data types
- ✓ Replace values
- ✓ Fill blanks

Shape

Structure for analysis

- ✓ Pivot / Unpivot
- ✓ Split columns
- ✓ Custom columns
- ✓ Extract values

Combine

Bring data together

- ✓ Merge queries
- ✓ Append queries
- ✓ Combine files
- ~ Validate after

Key questions before transforming

Business question

Final structure

Required columns

Data types

Duplicates and nulls

Query performance

"The goal is not to clean data randomly. The goal is to prepare it for the business question the report needs to answer."

Next up: Week 4 - Data Quality and Profiling

What is Power Query?

Power Query is the data preparation engine used in Power BI to connect, clean, shape, combine, and transform data before it is loaded into the model.

In simple terms: Power Query is where raw data becomes report-ready data.

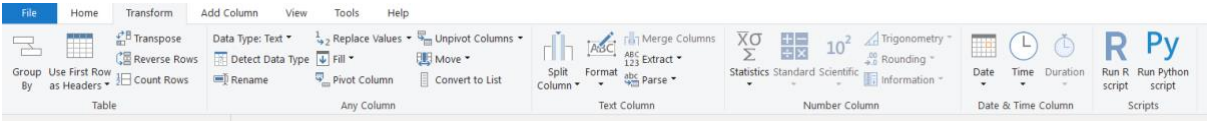
After connecting to a source, analysts use Power Query Editor to prepare the dataset for modelling and reporting. This includes removing unnecessary rows, changing data types, splitting columns, merging tables, appending data, unpivoting columns, creating custom columns, and fixing common data issues. The goal is not to clean data randomly. The goal is to prepare it for the business question the report needs to answer.

The important idea is this: Power BI reports are only as reliable as the data preparation behind them.

Pros	Cons
● Repeatable data cleaning steps ● Reduces manual Excel-based prep ● Improves consistency before modelling ● Transformations visible via Applied Steps ● Supports reusable preparation logic ● Reduces errors before reporting	■ Poorly planned steps slow queries ■ Too many steps become hard to maintain ■ Some transformations may break query folding ■ Incorrect cleaning can remove useful data ■ Complex logic needs stronger Power Query knowledge

Transform Menu: Overview

The Transform menu in Power Query gives analysts access to the main tools used to prepare data.



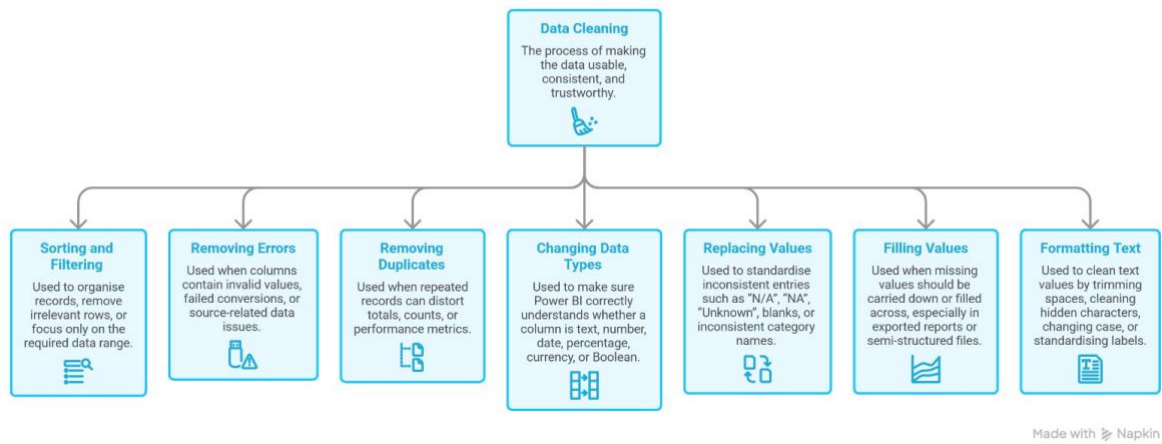
Power Query Transformation Components

Component	Description
 Table	Used for table-level operations such as grouping rows, using first row as headers, transposing data, reversing rows, and counting rows.
 Any Column	Used for general column operations such as detecting data types, renaming columns, replacing values, filling values, pivoting, unpivoting, moving columns, and converting values into lists.
 Text Column	Used when working with text-based fields. This includes splitting columns, formatting text, extracting values, parsing content, and merging text columns.
 Number Column	Used for numerical transformations such as statistics, rounding, standard calculations, scientific operations, and number-based adjustments.
 Date and Time Column	Used for date, time, and duration transformations. This helps analysts extract year, month, quarter, day, time, and duration-based values.
 Scripts	Used to run R or Python scripts where advanced data preparation or statistical logic is required.

Clean: Fixing the Data Before Analysis

Cleaning is the process of making the data usable, consistent, and trustworthy.

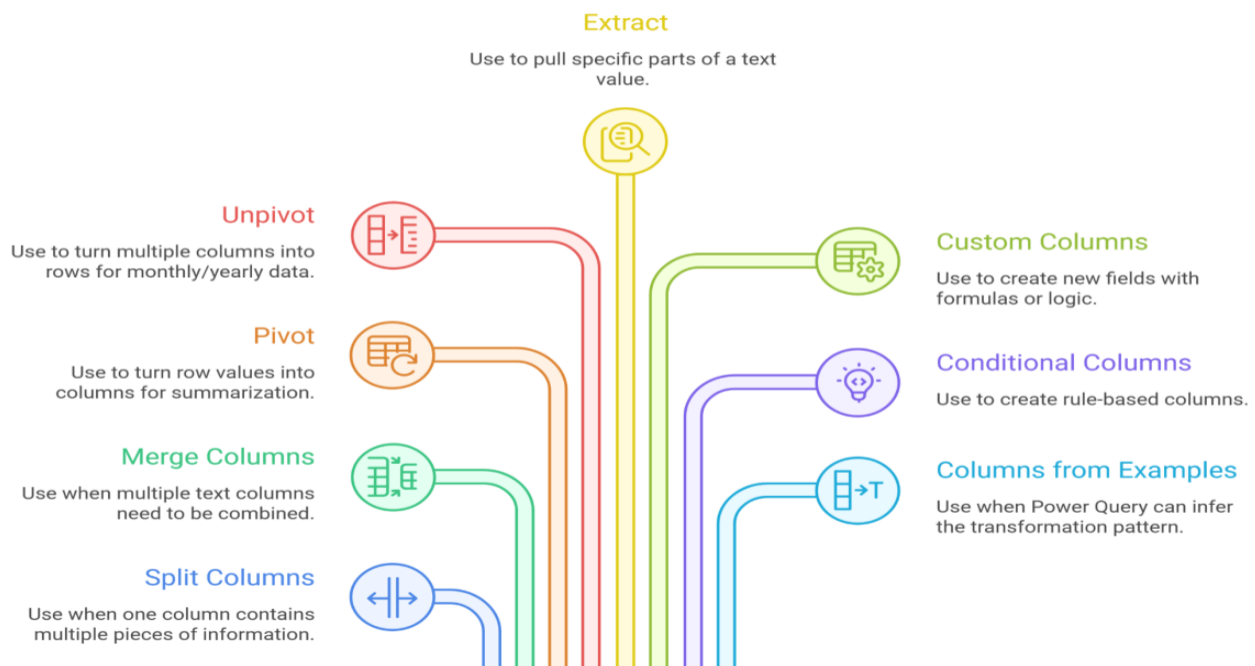
Clean: Fixing the Data Before Analysis



Shape: Structuring Data for Reporting

Shaping changes the structure of the dataset so it works better for modelling and analysis.

Shape: Structuring Data for Reporting



Made with Napkin

Combine and Transform: Bringing Data Together

Business data often sits across multiple tables, files, or systems. Power Query helps combine them into a usable structure.

Merge Queries: Used to join two tables together based on matching keys, similar to joins in SQL.

Example: Joining a Sales table with a Product table using Product ID.

Merge

Select a table and matching columns to create a merged table.

Mileage2018

State	Year	Miles (millions)
Alabama	2018	71167.20675
Alaska	2018	5487.319994
Arizona	2018	66144.51121
Arkansas	2018	36675.30905
California	2018	348795.7135

DrivingSafely2007-17

State	Year	Miles (millions)	Fatalities
Alabama	2007	61410	1110
Alaska	2007	5153	82
Arizona	2007	62963	1071
Arkansas	2007	33171	649
California	2007	328312	3995

Join Kind

Full Outer (all rows from both)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

OK

Cancel

Append Queries: Used to stack rows from similar tables into one table. **Example:** Combining January, February, and March sales tables into one sales dataset.

Append

Concatenate rows from two tables into a single table.

☒ Two tables ☐ Three or more tables

First table

Mileage2018

Second table

Fatalities2018

OK

Cancel

Combining Files: Used when multiple files in a folder have the same structure and need to be combined into a single table. **Example:** Combining monthly Excel files from a SharePoint folder.

Transforming After Combining: After merge, append, or combine operations, analysts often need to rename columns, change data types, remove extra fields, and validate the final structure.

Practical Way to Decide: What Transformation Do You Need?



A Simple Decision Guide for Power Query

Choose the right transformation based on the data issue you are solving.

1 INCORRECT VALUES		Use cleaning fixes when: The data contains wrong values, blanks, or formatting issues. Use Replace Values, Remove Errors, Fill, or Data Type changes.
2 UNNECESSARY DATA		Use row and column reduction when: The dataset includes fields or records you do not need. Use Remove Columns, Remove Rows, Filters, or Keep Rows.
3 POOR STRUCTURE		Use shaping tools when: The data layout is not analysis-ready. Use Split Columns, Merge Columns, Pivot, Unpivot, or Transpose.
4 MULTIPLE TABLES		Use Merge or Append when: You need to combine information from separate tables. Use Merge Queries for joins or Append Queries for stacking rows.
5 MULTIPLE SIMILAR FILES		Use Combine Files when: You receive repeated files with the same structure, such as monthly reports in a folder. Use Combine Files.
6 BUSINESS RULES		Use logic-based columns when: The transformation depends on business rules or derived fields. Use Custom Columns, Conditional Columns, or Columns From Examples.
7 ADVANCED LOGIC		Use scripting when: The transformation requires advanced or custom logic. Use Power Query M, Python, or R scripts.

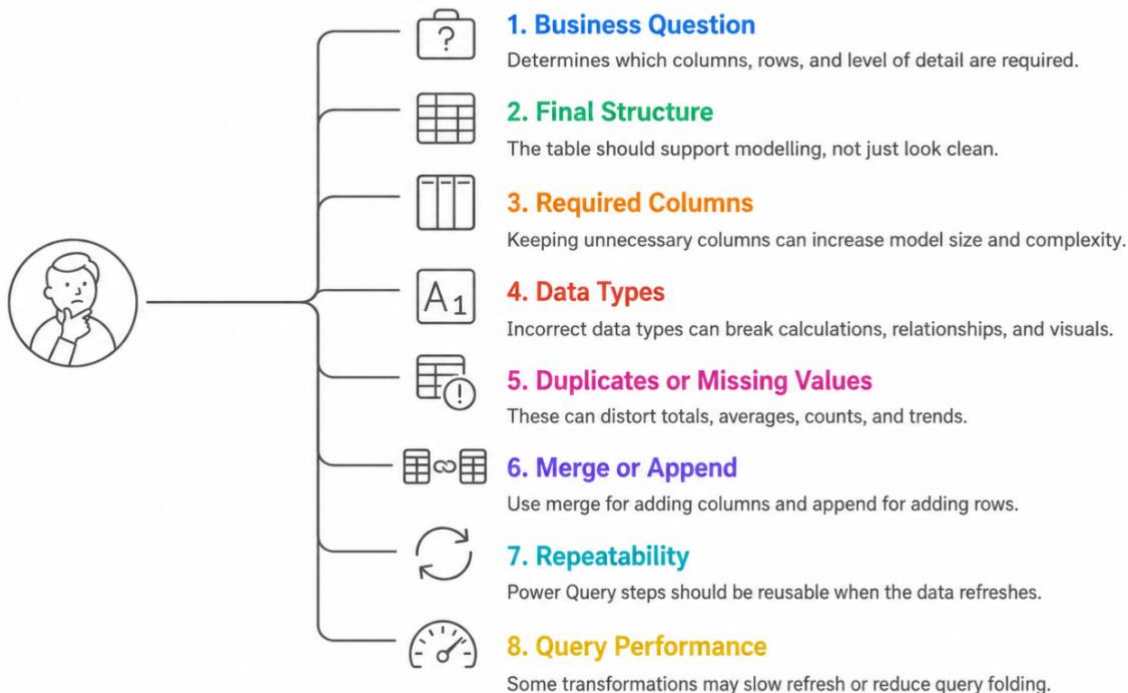


Power Query helps turn raw data into analysis-ready tables.

What Analysts Should Ask Before Cleaning the Data

What Analysts Should Ask Before Cleaning the Data

Key questions to ask before applying Power Query transformations



Conclusion

Power Query is where data preparation becomes visible, repeatable, and structured. It allows analysts to clean messy data, shape it for analysis, combine multiple sources, and create transformation steps that refresh consistently over time. Good Power BI reports do not rely on perfect source data. They rely on analysts knowing how to prepare imperfect data properly.

Stay tuned for next week's article in **The Insight Stack: Power BI, Layer by Layer**, where we move into **Week 4: Data Quality** and explore how analysts check whether data is complete, consistent, and reliable before deeper modelling and reporting.

Further Reading

- [Using Power Query to transform data](#)
- [Applied Steps in Power Query](#)
- [Using the data profiling tools](#)
- [Using the Queries pane](#)
- [Choose or remove columns](#)
- [Add a custom column](#)
- [Fill values in a column](#)
- [Append queries](#)
- [Merge queries overview](#)
- [Split columns by positions](#)